Hormonal therapy for prostate cancer

Hormonal therapies lower testosterone levels or stop testosterone reaching the prostate cancer cells. Hormonal therapy may be given on its own, or in combination with other treatments.

On this page

- What is hormonal therapy for prostate cancer?
- Hormonal therapy with radiotherapy
- Hormonal therapy on its own
- Types of hormonal therapy
- Side effects of hormonal therapy for prostate cancer
- How we can help

What is hormonal therapy for prostate cancer?

Hormonal therapies work by altering the production or activity of particular hormones in the body.

Prostate cancer needs the hormone testosterone to grow. Testosterone is mainly made by the testicles. Some hormonal therapies for prostate cancer reduce the amount of testosterone naturally produced in the body. Others stop testosterone from reaching the prostate cancer cells.

Testosterone is important for:

- sex drive (libido)
- getting an erection
- facial and body hair
- muscle development and bone strength
- heart health.

Hormonal therapy with radiotherapy

Doctors often advise having hormonal therapy with radiotherapy (including brachytherapy). They usually recommend it if you have intermediate risk or high risk early prostate cancer, or to treat locally advanced prostate cancer.

Hormonal therapy can shrink the cancer that helps make radiotherapy more effective. Your cancer doctor may advise you to have hormonal therapy for 3 to 6 months in the time before, during and after radiotherapy. Or you may be advised to have hormonal therapy over a longer period, up to 3 years.

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Your cancer doctor can tell you how long is suitable for you.

Abiraterone

In addition to the longer hormonal therapy, some people may also have another hormonal therapy called abiraterone for 2 years. Your cancer doctor can talk to you about whether this is an option for you.

Until recently, abiraterone only has been used to treat prostate cancer that has spread to other parts of the body. This is called advanced or metastatic prostate cancer.

Clinical trials have now shown that abiraterone may be helpful in treating high risk prostate cancer that has not yet spread to other parts of the body (non-metastatic cancer).

Abiraterone tablets are usually taken once a day. They should not be taken with food as it can affect how abiraterone works and increase side effects.

Abiraterone is taken with a daily steroid tablet. The steroids help reduce some of the side effects of abiraterone.

Hormonal therapy on its own

Doctors do not usually advise having hormonal therapy instead of surgery or radiotherapy. Hormonal therapy alone cannot cure the cancer. But depending on your general health and preferences, you may decide to have hormonal therapy on its own. For example, you may decide this if you:

- are not well enough to have surgery or radiotherapy
- do not want to have surgery or radiotherapy
- are having watchful waiting and the cancer starts to grow.

Hormonal therapy can slow down or stop the cancer cells growing for many years. It can improve the symptoms caused by the cancer. Not having radiotherapy means you avoid its side effects, such as bladder and bowel effects. But hormonal therapy can also cause side effects.

It is important to talk to your cancer doctor or specialist nurse before you decide.

Intermittent hormonal therapy for locally advanced prostate cancer

Intermittent hormonal therapy may sometimes be an option for locally advanced prostate cancer. This is when you stop taking hormonal therapy tablets for a time, then start taking them again after a while. This is to give you a break from the side effects of hormonal therapy.

Intermittent hormonal therapy is not suitable for everyone and should only be done following your cancer doctor’s advice. Your cancer doctor can explain more about this. They usually measure your PSA level every 3 months. If your PSA goes up to a certain level or you get symptoms, your doctor may advise you to start having hormonal therapy again.

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Types of hormonal therapy

There are different types of hormonal therapy. Your cancer doctor, specialist nurse or pharmacist will explain the treatment that is most suitable for you.

LHRH agonists

The pituitary gland in the brain makes a hormone called luteinising hormone (LH). This hormone tells the testicles to make testosterone.

LH-releasing hormone (LHRH) agonists interfere with this action and stop the testicles making testosterone. You have them as an implant injection or an injection under the skin.

The commonly used LHRH agonists are:

- goserelin (Zoladex®, Zoladex LA®)
- leuprorelin
- triptorelin (Decapeptyl®, Gonapeptyl®).

Your practice nurse at your GP surgery can give you these drugs. If you are not able to visit the GP surgery, a district nurse may give you the injections at home. Or they may be given by a nurse at the hospital.

You have goserelin as an injection of a small pellet (implant) under the skin of your tummy (abdomen). The drug is released slowly as the pellet dissolves. You have it every 4 weeks, or as a longer-acting injection every 12 weeks.

Leuprorelin and triptorelin are given as an injection under the skin (subcutaneously) or into a muscle (intramuscularly). You have these monthly, or every 3 or 6 months. Your cancer doctor, specialist nurse or pharmacist will talk to you about your treatment plan.

The first time you have one of these drugs, it can cause a temporary increase in testosterone. This can make any symptoms worse for a short time. This is sometimes called tumour flare. To prevent this, your doctor usually asks you to take an anti-androgen drug, such as bicalutamide. You take it for a short time before and after starting the LHRH agonist.

GnRH antagonists

Gonadotropin-releasing hormone (GnRH) antagonists block messages from the brain to the testicles that tell them to make testosterone. They do not cause tumour flare, so you do not need to take an anti-androgen with a GnRH antagonist.

Some people may have a GnRH antagonist called relugolix (Orgovyx®). Relugolix is taken as tablets. This drug may be used if you have other health conditions.

Anti-androgen drugs

An anti-androgen drug stops testosterone from reaching the cancer cells. This is usually a drug called bicalutamide. You take bicalutamide as tablets. You may have it with radiotherapy, instead of having an LHRH agonist with radiotherapy. Or you may have bicalutamide before and after the first injection of

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an LHRH agonist. This is to prevent any symptoms getting temporarily worse – for example, tumour flare.

Side effects of hormonal therapy for prostate cancer

Reducing the level of testosterone can cause different side effects. There are different ways hormonal side effects can be managed or treated. Your cancer doctor, specialist nurse or pharmacist will explain this to you. Some side effects are only likely to affect you when you have hormonal therapy for more than 6 months.

Different hormonal therapies have different side effects. It is important to discuss these with your cancer doctor, specialist nurse or pharmacist before treatment, so you know what to expect.

Common side effects of hormonal therapy

Common side effects include:

Erection difficulties and reduced sex drive

Hormonal therapy may make it difficult for you to get or keep an erection. This is called erectile dysfunction (ED). This will usually improve slowly over a few months after you stop hormonal treatment, depending on how long you take it for. Hormonal therapy may also lower your desire to have sex (libido).

Hormonal therapy can also cause some changes to the penis – for example, it may appear shorter. Your cancer team can talk to you about ways of preventing or helping this.

If you have ED, there are drugs and treatments that may help. Even with a low sex drive, some ED treatments may work for you.

Hot flushes and sweats

Hormonal therapy can cause hot flushes. These may reduce as your body adjusts to hormonal treatment. In most cases, they gradually improve after treatment finishes. Talk to your cancer doctor or specialist nurse if you are having problems. They can give you advice and may be able to prescribe medicines to help.

Tiredness and difficulty sleeping

Feeling tired is a very common side effect of hormonal therapy. Regular physical activity, such as walking, can help reduce tiredness. Physical activity can also help reduce other side effects from hormonal therapy. Try to keep to your regular routine if you can.

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Hot flushes may make sleeping difficult, so managing them may help you to sleep better.

Mood changes

During hormonal therapy, you may have mood swings, or feel low or depressed. Talking to family and friends about how you feel might help. If mood changes last for more than a few weeks, tell your GP, cancer doctor or specialist nurse. They can talk to you about different ways to manage low mood or depression.

Memory and concentration problems

Problems with memory and concentration may be caused by the hormonal therapy, or because of tiredness or feeling anxious.

Other side effects of hormonal therapy

If you have hormonal therapy for 6 months or more, you may have other side effects. The benefits of hormonal therapy generally outweigh the possible risks. Your cancer doctor, specialist nurse or pharmacist will talk to you about this.

Other possible side effects include:

Weight gain and loss of muscle strength

You may gain weight (especially around the middle) and lose muscle strength. Regular physical activity and a healthy, balanced diet can help manage this.

Breast swelling or tenderness

Breast swelling or tenderness is more common if you have flutamide or bicalutamide over a longer period. You may be given low dose radiotherapy to your chest before treatment to prevent breast swelling. If you are taking bicalutamide, another option is to take a hormonal drug called tamoxifen to reduce breast swelling.

Bone thinning (osteoporosis)

The risk of osteoporosis is increased with long term hormonal therapy. You may have a scan to check your bones before you start treatment. Regular weight-bearing exercises such as walking, dancing, hiking, or gentle weightlifting can help keep your bones healthy. Your cancer doctor or specialist nurse may give you advice on diet and exercise.

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Your cancer doctor, specialist nurse or pharmacist may advise you to take calcium and vitamin D tablets. Depending on your bone health, your cancer doctor may talk to you about bone-strengthening drugs called bisphosphonates.

Increased risk of heart disease and diabetes

Not smoking, being physically active, eating healthily and keeping to a healthy weight can help reduce the risk of heart disease and diabetes.

About our information

This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer.

References

Below is a sample of the sources used in our prostate cancer information. If you would like more information about the sources we use, please contact us at informationproductionteam@macmillan.org.uk

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Our cancer information meets the PIF TICK quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about how we produce our information.

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as ‘you’ so that everyone feels included. Where clinically necessary we use the terms ‘men’ and ‘women’ or ‘male’ and ‘female’. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information here.

How we can help

Clinical Information Nurse Specialists

Our Cancer Information Nurse Specialists are dedicated cancer nurses available to talk to on our Macmillan Cancer Support Line.

Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.

Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don’t want to call.

Online Community

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Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our

Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

Get help

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